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Commercial National Security Algorithm (CNSA) Suite Cryptography for Internet Protocol Security (IPsec)

Abstract

The United States Government has published the National Security Agency's Commercial National Security Algorithm (CNSA) Suite, which defines cryptographic algorithm policy for national security applications. This document specifies the conventions for using the United States National Security Agency's CNSA Suite algorithms in Internet Protocol Security (IPsec). It applies to the capabilities, configuration, and operation of all components of US National Security Systems (described in NIST Special Publication 800-59) that employ IPsec. This document is also appropriate for all other US Government systems that process high-value information. It is made publicly available for use by developers and operators of these and any other system deployments.

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Table of Contents

1. Introduction
2. Terminology
3. The Commercial National Security Algorithm Suite
4. CNSA-Compliant IPsec Overview
5. IPsec User Interface Suites
 - 5.1. Suite CNSA-GCM-256-ECDH-384
 - 5.2. Suite CNSA-GCM-256-DH-3072
 - 5.3. Suite CNSA-GCM-256-DH-4096
6. IKEv2 Authentication
7. Certificates
8. IKEv2 Security Associations (SAs)
9. The Key Exchange Payload in the IKE_SA_INIT Exchange
10. Generating Key Material for the IKE SA
11. Additional Requirements
12. Guidance for Applications with Long Data-Protection Requirements
13. Security Considerations
14. IANA Considerations
15. References
 - 15.1. Normative References
 - 15.2. Informative References

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1. Introduction

This document specifies the conventions for using the United States National Security Agency's (NSA's) Commercial National Security Algorithm (CNSA) Suite algorithms [[CNSA](#)] in Internet Protocol Security (IPsec). It defines CNSA-based User Interface suites ("UI suites") describing sets of security configurations for Internet Key Exchange Protocol Version 2 (IKEv2) and IP Encapsulating Security Payload (ESP) protocol use, and specifies certain other constraints with respect to algorithm selection and configuration. It applies to the capabilities, configuration, and operation of all components of US National Security Systems (described in NIST Special Publication 800-59 [[SP80059](#)]) that employ IPsec. This document is also appropriate for all other US Government systems that process high-value information. It is made publicly available for use by developers and operators of these and any other system deployments.

The reader is assumed to have familiarity with the following:

- "IP Encapsulating Security Payload (ESP)" [[RFC4303](#)]

- "Internet X.509 Public Key Infrastructure Certificate and Certificate Revocation List (CRL) Profile" [RFC5280]
- "Internet Key Exchange Protocol Version 2 (IKEv2)" [RFC7296]
- "Cryptographic Algorithm Implementation Requirements and Usage Guidance for Encapsulating Security Payload (ESP) and Authentication Header (AH)" [RFC8221]
- "Commercial National Security Algorithm (CNSA) Suite Certificate and Certificate Revocation List (CRL) Profile" [RFC8603]

2. Terminology

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

AES refers to the Advanced Encryption Standard. ECDSA and ECDH refer to the use of the Elliptic Curve Digital Signature Algorithm (ECDSA) and Elliptic Curve Diffie-Hellman (ECDH), respectively. DH refers to Diffie-Hellman key establishment. RSA refers to an RSA signature.

3. The Commercial National Security Algorithm Suite

The NSA profiles commercial cryptographic algorithms and protocols as part of its mission to support secure, interoperable communications for US Government National Security Systems. To this end, it publishes guidance to both (1) assist with the US Government transition to new algorithms and (2) provide vendors -- and the Internet community in general -- with information concerning their proper use and configuration.

Recently, cryptographic transition plans have become overshadowed by the prospect of the development of a cryptographically relevant quantum computer. The NSA has established the Commercial National Security Algorithm (CNSA) Suite to provide vendors and IT users near-term flexibility in meeting their information assurance interoperability requirements. The purpose behind this flexibility is to avoid vendors and customers making two major transitions in a relatively short timeframe, as we anticipate a need to shift to quantum-resistant cryptography in the near future.

The NSA is authoring a set of RFCs, including this one, to provide updated guidance concerning the use of certain commonly available commercial algorithms in IETF protocols. These RFCs can be used in conjunction with other RFCs and cryptographic guidance (e.g., NIST Special Publications) to properly protect Internet traffic and data-at-rest for US Government National Security Systems.

4. CNSA-Compliant IPsec Overview

CNSA-compliant implementations for IPsec **MUST** use IKEv2 [RFC7296].

Implementing a CNSA-compliant IPsec system requires cryptographic algorithm selection for both the IKEv2 and ESP protocols. The following CNSA requirements apply to IPsec:

Encryption:

AES [FIPS197] (with key size 256 bits)

Digital Signature:

ECDSA [FIPS186] (using the NIST P-384 elliptic curve)

RSA [FIPS186] (with a modulus of 3072 bits or larger)

Key Establishment:

ECDH [SP80056A] (using the NIST P-384 elliptic curve)

DH [RFC3526] (with a prime modulus of 3072 or larger)

To facilitate selection of appropriate combinations of compliant algorithms, this document provides CNSA-compliant User Interface suites (UI suites) [RFC4308] to implement IPsec on National Security Systems.

The approved CNSA hash function for all purposes is SHA-384, as defined in [FIPS180]. However, PRF_HMAC_SHA_512 is specified for the IKEv2 Pseudorandom Function (PRF) instead of PRF_HMAC_SHA_384, due to availability. See Section 8 below.

For CNSA Suite applications, public key certificates **MUST** be compliant with the CNSA Suite Certificate and CRL Profile specified in [RFC8603].

Under certain conditions, such as applications having long-lived data-protection requirements, systems that do not comply with the requirements of this document are acceptable; see Section 12.

5. IPsec User Interface Suites

User Interface (UI) suites [RFC4308] are named suites that cover some typical security policy options for IPsec. Use of UI suites does not change the IPsec protocol in any way. The following UI suites provide cryptographic algorithm choices for ESP [RFC4303] and for IKEv2 [RFC7296]. The selection of a UI suite will depend on the key exchange algorithm. The suite names indicate the Advanced Encryption Standard [FIPS197] mode, AES key length specified for encryption, and the key exchange algorithm.

Although RSA is also a CNSA-approved key establishment algorithm, only DH and ECDH are specified for key exchange in IKEv2 [RFC7296]. RSA in IPsec is used only for digital signatures. See Section 6.

ESP requires negotiation of both a confidentiality algorithm and an integrity algorithm. However, algorithms for Authenticated Encryption with Associated Data (AEAD) [RFC5116] do not require a separate integrity algorithm to be negotiated. In particular, since AES-GCM is an AEAD algorithm, ESP implementing AES-GCM **MUST** either offer no integrity algorithm or indicate the single integrity algorithm NONE (see Section 3.3 of [RFC7296]).

To be CNSA compliant, IPsec implementations that use the following UI suites **MUST** use the suite names listed below. IPsec implementations **SHOULD NOT** use names different than those listed here for the suites that are described and **MUST NOT** use the names listed here for suites that do not match these values. These requirements are necessary for interoperability.

Transform names are as listed in the IANA "Internet Key Exchange Version 2 (IKEv2) Parameters" registry. Definitions of the transforms are contained in the references specified in that registry.

Other UI suites may be acceptable for CNSA compliance. See Section 8 for details.

5.1. Suite CNSA-GCM-256-ECDH-384

ESP SA:

Encryption: ENCR_AES_GCM_16 (with key size 256 bits)

Integrity: NONE

IKEv2 SA:

Encryption: ENCR_AES_GCM_16 (with key size 256 bits)

PRF: PRF_HMAC_SHA2_512

Integrity: NONE

Diffie-Hellman group: 384-bit random ECP group

5.2. Suite CNSA-GCM-256-DH-3072

ESP SA:

Encryption: ENCR_AES_GCM_16 (with key size 256 bits)

Integrity: NONE

IKEv2 SA:

Encryption: ENCR_AES_GCM_16 (with key size 256 bits)

PRF: PRF_HMAC_SHA2_512

Integrity: NONE

Diffie-Hellman group: 3072-bit MODP group

5.3. Suite CNSA-GCM-256-DH-4096

ESP SA:

Encryption: ENCR_AES_GCM_16 (with key size 256 bits)

Integrity: NONE

IKEv2 SA:

Encryption: ENCR_AES_GCM_16 (with key size 256 bits)

PRF: PRF_HMAC_SHA2_512

Integrity: NONE

Diffie-Hellman group: 4096-bit MODP group

6. IKEv2 Authentication

Authentication of the IKEv2 Security Association (SA) requires computation of a digital signature. To be CNSA compliant, digital signatures **MUST** be generated with SHA-384 as defined in [\[RFC8017\]](#) together with either ECDSA-384 as defined in [\[RFC4754\]](#) or RSA with 3072-bit or greater modulus. (For applications with long data-protection requirements, somewhat different rules apply; see [Section 12](#).)

Initiators and responders **MUST** be able to verify ECDSA-384 signatures and **MUST** be able to verify RSA with 3072-bit or 4096-bit modulus and SHA-384 signatures.

For CNSA-compliant systems, authentication methods other than ECDSA-384 or RSA **MUST NOT** be accepted for IKEv2 authentication. A 3072-bit modulus or larger **MUST** be used for RSA. If the relying party receives a message signed with any authentication method other than an ECDSA-384 or RSA signature, it **MUST** return an AUTHENTICATION_FAILED notification and stop processing the message. If the relying party receives a message signed with RSA using less than a 3072-bit modulus, it **MUST** return an AUTHENTICATION_FAILED notification and stop processing the message.

7. Certificates

To be CNSA compliant, the initiator and responder **MUST** use X.509 certificates that comply with [\[RFC8603\]](#). Peer authentication decisions must be based on the Subject or Subject Alternative Name from the certificate that contains the key used to validate the signature in the Authentication Payload as defined in [Section 3.8](#) of [\[RFC7296\]](#), rather than the Identification Data from the Identification Payload that is used to look up policy.

8. IKEv2 Security Associations (SAs)

[Section 5](#) specifies three UI suites for ESP and IKEv2 Security Associations. All three use AES-GCM for encryption but differ in the key exchange algorithm. Galois/Counter Mode (GCM) [[RFC4106](#)] combines counter (CTR) mode with a secure, parallelizable, and efficient authentication mechanism. Since AES-GCM is an AEAD algorithm, ESP implements AES-GCM with no additional integrity algorithm (see [Section 3.3](#) of [[RFC7296](#)]).

An initiator proposal **SHOULD** be constructed from one or more of the following suites:

- CNSA-GCM-256-ECDH-384
- CNSA-GCM-256-DH-3072
- CNSA-GCM-256-DH-4096

A responder **SHOULD** accept proposals constructed from at least one of the three named suites. Other UI suites may result in acceptable proposals (such as those based on PRF_HMAC_SHA2_384); however, these are provided to promote interoperability.

Nonce construction for AES-GCM using a misuse-resistant technique [[RFC8452](#)] conforms with the requirements of this document and **MAY** be used if a Federal Information Processing Standard (FIPS) validated implementation is available.

The named UI suites specify PRF_HMAC_SHA2_512 for the IKEv2 SA PRF transform, as PRF_HMAC_SHA2_384 is not listed among required PRF transforms in [[RFC8247](#)]; therefore, implementation of the latter is likely to be scarce. The security level of PRF_HMAC_SHA2_512 is comparable, and the difference in performance is negligible. However, SHA-384 is the official CNSA algorithm for computing a condensed representation of information. Therefore, the PRF_HMAC_SHA2_384 transform is CNSA compliant if it is available to the initiator and responder. Any PRF transform other than PRF_HMAC_SHA2_384 or PRF_HMAC_SHA2_512 **MUST NOT** be used.

If none of the proposals offered by the initiator consist solely of transforms based on CNSA algorithms (such as those in the UI suites defined in [Section 5](#)), the responder **MUST** return a Notify payload with the error NO_PROPOSAL_CHOSEN when operating in CNSA-compliant mode.

9. The Key Exchange Payload in the IKE_SA_INIT Exchange

The key exchange payload is used to exchange Diffie-Hellman public numbers as part of a Diffie-Hellman key exchange. The CNSA-compliant initiator and responder **MUST** each generate an ephemeral key pair to be used in the key exchange.

If the Elliptic Curve Diffie-Hellman (ECDH) key exchange is selected for the SA, the initiator and responder both **MUST** generate an elliptic curve (EC) key pair using the P-384 elliptic curve. The ephemeral public keys **MUST** be stored in the key exchange payload as described in [[RFC5903](#)].

If the Diffie-Hellman (DH) key exchange is selected for the SA, the initiator and responder both **MUST** generate a key pair using the appropriately sized MODP group as described in [[RFC3526](#)]. The size of the MODP group will be determined by the selection of either a 3072-bit or greater modulus for the SA.

10. Generating Key Material for the IKE SA

As noted in [Section 7](#) of [[RFC5903](#)], the shared secret result of an ECDH key exchange is the 384-bit x value of the ECDH common value. The shared secret result of a DH key exchange is the number of octets needed to accommodate the prime (e.g., 384 octets for 3072-bit MODP group) with leading zeros as necessary, as described in [Section 2.1.2](#) of [[RFC2631](#)].

IKEv2 allows for the reuse of Diffie-Hellman private keys; see [Section 2.12](#) of [\[RFC7296\]](#). However, there are security concerns related to this practice. Section 5.6.3.3 of [\[SP80056A\]](#) states that an ephemeral private key **MUST** be used in exactly one key establishment transaction and **MUST** be destroyed (zeroized) as soon as possible. Section 5.8 of [\[SP80056A\]](#) states that any shared secret derived from key establishment **MUST** be destroyed (zeroized) immediately after its use. CNSA-compliant IPsec systems **MUST** follow the mandates in [\[SP80056A\]](#).

11. Additional Requirements

The IPsec protocol AH **MUST NOT** be used in CNSA-compliant implementations.

A Diffie-Hellman group **MAY** be negotiated for a Child SA as described in [Section 1.3](#) of [\[RFC7296\]](#), allowing peers to employ Diffie-Hellman in the CREATE_CHILD_SA exchange. If a transform of type 4 is specified for an SA for ESP, the value of that transform **MUST** match the value of the transform used by the IKEv2 SA.

Per [\[RFC7296\]](#), if a CREATE_CHILD_SA exchange includes a KEi payload, at least one of the SA offers **MUST** include the Diffie-Hellman group of the KEi. For CNSA-compliant IPsec implementations, the Diffie-Hellman group of the KEi **MUST** use the same group used in the IKE_INIT_SA.

For IKEv2, rekeying of the CREATE_CHILD_SA **MUST** be supported by both parties. The initiator of this exchange **MAY** include a new Diffie-Hellman key; if it is included, it **MUST** use the same group used in the IKE_INIT_SA. If the initiator of the exchange includes a Diffie-Hellman key, the responder **MUST** include a Diffie-Hellman key, and it **MUST** use the same group.

For CNSA-compliant systems, the IKEv2 authentication method **MUST** use an end-entity certificate provided by the authenticating party. Identification Payloads (IDi and IDr) in the IKE_AUTH exchanges **MUST NOT** be used for the IKEv2 authentication method but may be used for policy lookup.

The administrative User Interface (UI) for a system that conforms to this profile **MUST** allow the operator to specify a single suite. If only one suite is specified in the administrative UI, the IKEv2 implementation **MUST** only offer algorithms for that one suite.

The administrative UI **MAY** allow the operator to specify more than one suite; if it allows this, it **MUST** allow the operator to specify a preferred order for the suites that are to be offered or accepted. If more than one suite is specified in the administrative UI, the IKEv2 implementation **MUST** only offer algorithms of those suites. (Note that although this document does not define a UI suite specifying PRF_HMAC_SHA2_384, a proposal containing such a transform is CNSA compliant.)

12. Guidance for Applications with Long Data-Protection Requirements

The CNSA mandate is to continue to use current algorithms with increased security parameters, then transition to approved post-quantum resilient algorithms when they are identified. However, some applications have data-in-transit-protection requirements that are long enough that post-quantum resilient protection must be provided now. Lacking approved asymmetric post-quantum resilient confidentiality algorithms, that means approved symmetric techniques must be used as described in this section until approved post-quantum resilient asymmetric algorithms are identified.

For new applications, confidentiality and integrity requirements from the sections above **MUST** be followed, with the addition of a Pre-Shared Key (PSK) mixed in as defined in [\[RFC8784\]](#).

Installations currently using IKEv1 with PSKs **MUST** (1) use AES in cipher block chaining mode (AES-CBC) in conjunction with a CNSA-compliant integrity algorithm (e.g., AUTH_HMAC_SHA2_384_192) and (2) transition to IKEv2 with PSKs [\[RFC8784\]](#) as soon as implementations become available.

Specific guidance for systems not compliant with the requirements of this document, including non-GCM modes and PSK length, and PSK randomness, will be defined in solution-specific requirements appropriate to the application. Details of those requirements will depend on the program under which the commercial National Security Systems solution is developed (e.g., an NSA Commercial Solutions for Classified Capability Package).

13. Security Considerations

This document inherits all of the security considerations of the IPsec and IKEv2 documents, including [RFC7296], [RFC4303], [RFC4754], and [RFC8221].

The security of a system that uses cryptography depends on both the strength of the cryptographic algorithms chosen and the strength of the keys used with those algorithms. The security also depends on the engineering and administration of the protocol used by the system to ensure that there are no non-cryptographic ways to bypass the security of the overall system.

When selecting a mode for the AES encryption [RFC5116], be aware that nonce reuse can result in a loss of confidentiality. Nonce reuse is catastrophic for GCM, since it also results in a loss of integrity.

14. IANA Considerations

IANA has added the UI suites defined in this document to the "Cryptographic Suites for IKEv1, IKEv2, and IPsec" registry located at <<https://www.iana.org/assignments/crypto-suites>>:

Identifier	Reference
CNSA-GCM-256-ECDH-384	RFC 9206
CNSA-GCM-256-DH-3072	RFC 9206
CNSA-GCM-256-DH-4096	RFC 9206

Table 1

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